

Safety Checklists

Device inspection

- ☐ Bank voltage reads 0 V on the connected multimeter.
- ☐ Busbars show no cracking along the flattened edges and have no loosened joints.
- ☐ Nothing is touching the busbars except at the intended connection sites.
- ☐ Bleeders and voltage divider resistors are visually intact and connected at their respective terminals.
- ☐ Safety discharge stick is connected to the bank-negative busbar through discharge resistors and is within reach.
- ☐ Correct transformer tap is selected for the target voltage.
- ☐ Transformer taps are far away from each other.
- ☐ Target voltage does not exceed the system's maximum voltage.
- ☐ Enclosure lid is closed; lid interlock closed.
- ☐ Dead-man interlock is held by the second person.
- ☐ Comparator trip threshold is set for the intended voltage.
- ☐ Multimeter is readable and enclosed with the high voltage circuitry.
- ☐ PSU current limit set according to expected maximum current draw.
- ☐ TUI divider ratio matches the divider tap actually in use.
- ☐ TUI shows a nominal state. Press R to reset if needed.

Pre-charge safety checklist

- ☐ At least two people are present. Both have been briefed on the location and operation of the charge-cutoff switch, the discharge stick, and the de-energization checklist.
- ☐ Operator is wearing Class 1 insulating rubber gloves and insulated footwear. Every person present is wearing impact-rated eye protection.
- ☐ No rings, watches, or other conductive materials on either person.
- ☐ Bench is clear of conductive clutter, tools, and stray wires.
- ☐ No liquids on or near the setup.
- ☐ A physical barrier separates the operator from the high-voltage components.
- ☐ Both people are clear of the high-voltage region and ready for the circuit to be energized.

De-energization

- ☐ Stop charging by pressing C again or pressing A to arm the system. The current draw should drop to quiescent levels.
- ☐ Cut the PSU feed either by releasing the dead-man interlock or by pressing the charge-cutoff switch.
- ☐ Turn the PSU off with its respective on/off switch, or by disconnecting the PSU directly from the wall.
- ☐ Verify the multimeter is reading a non-zero voltage. Be wary of zero voltage readings as the multimeter might have been disconnected from the circuit or might be malfunctioning.
- ☐ Allow the bleeders to drain the bank (leave untouched for at least 20 minutes), or
- ☐ Make firm, quick contact with the safety discharge stick hook on the bank positive terminal through the enclosure port. Hold contact for at least 5 seconds.
- ☐ Verify the multimeter reads a bank voltage of 0 V. Verify the multimeter probes are connected to the bank.
- ☐ If the probes of the multimeter appear disconnected, connect the discharge stick for another 10 seconds and leave the device untouched for at least 20 minutes.